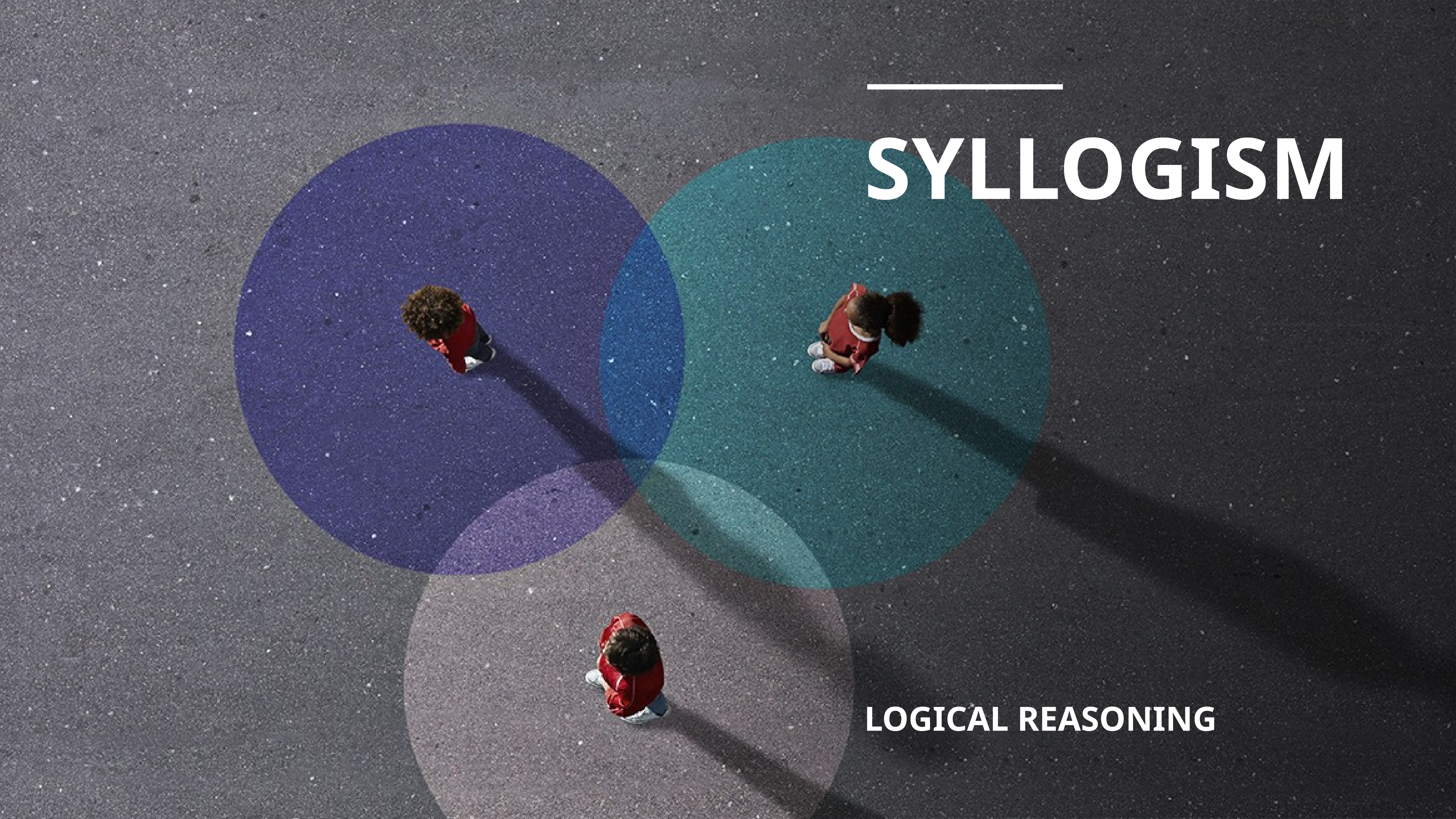


SYLLOGISM

LOGICAL REASONING

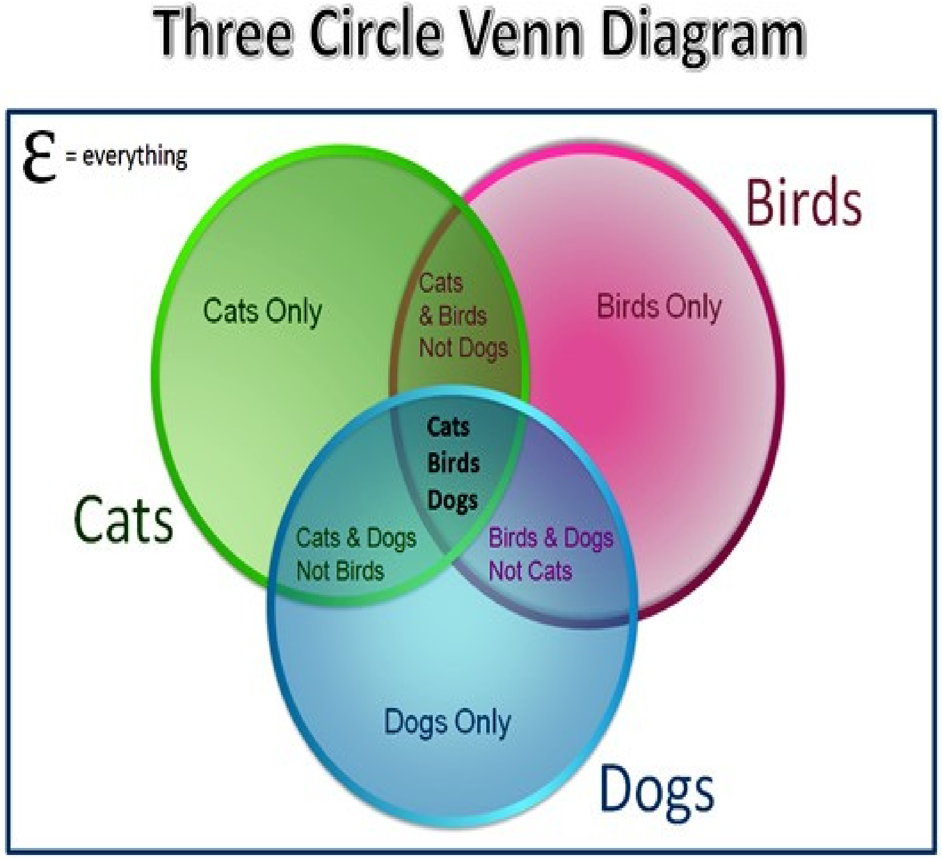
SYLLOGISMS/DEDUCTIONS

- Syllogism comes from the Greek word ***"syllogismos"***, meaning **conclusion or inference**. It involves drawing conclusions from given statements using deductive reasoning.
- Questions in this section present **two or more statements**, followed by **two or more conclusions**. The task is to determine which **conclusions logically follow**, assuming all **statements are true, even if they contradict common facts.**
- The most effective method for solving syllogisms is using **Venn diagrams**. One should draw all possible diagrams based on the statements, **analyze** each, and select the conclusion(s) common to all diagrams as the correct answer.

VENN DIAGRAM

- **Venn diagrams** are **pictorial representations** of sets using geometrical shapes. A **combined diagram** is used to represent all given statements.

- Each set of statements can be represented in **multiple valid ways** using Venn diagrams. A conclusion is said to **definitely follow** only if it holds true in **all possible diagrams**.



QUALIFIERS IN SYLLOGISM

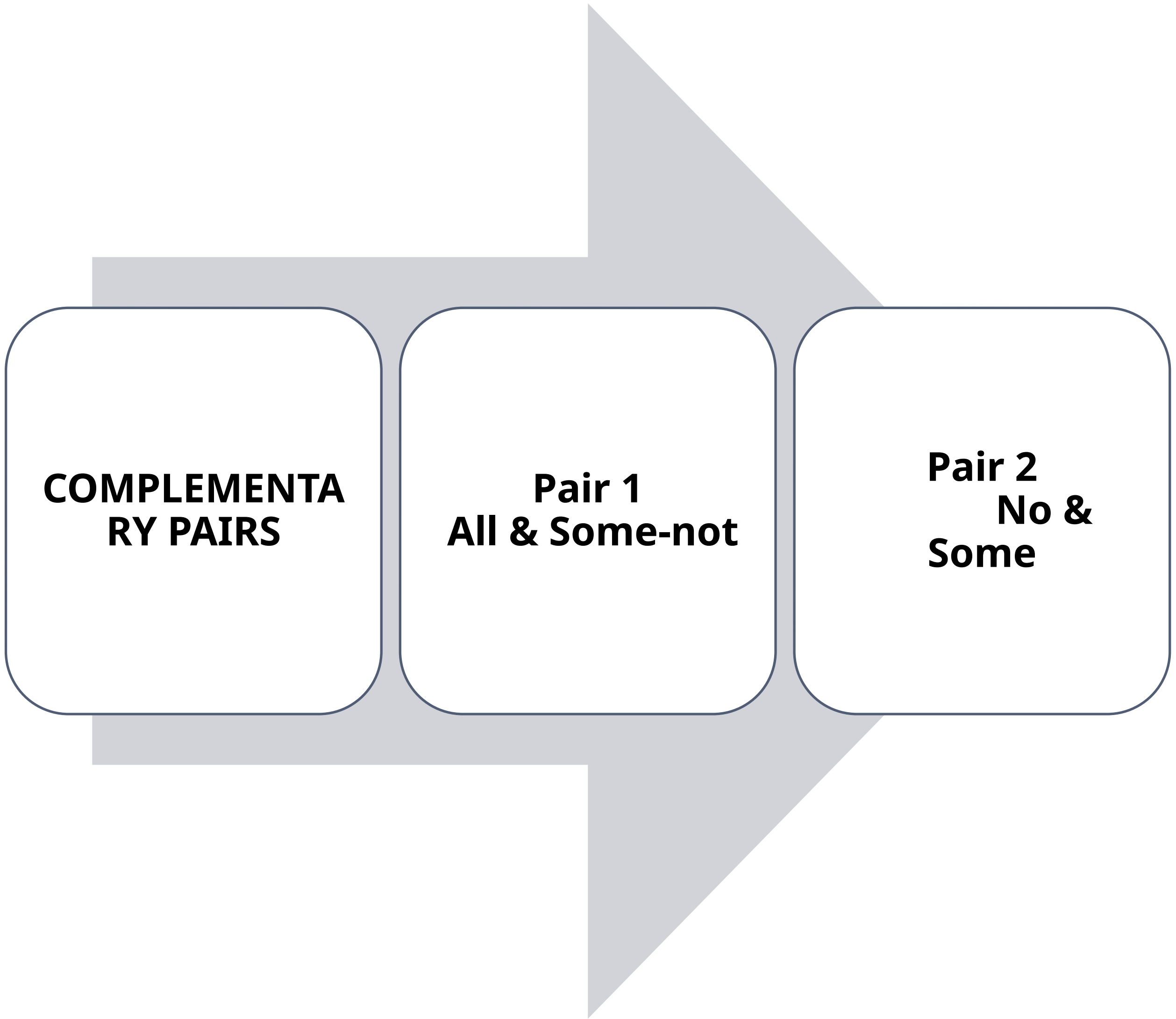
Qualifier	Form	Example
All	All A are B	All animals are living things
No	No A is B	No boy is a girl
Some	Some A are B	Some doctors are men
Some Not	Some A are not B	Some cricketers are not Indians

EQUIVALENT PHRACES

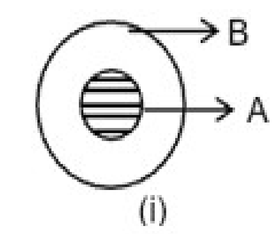
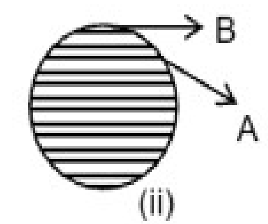
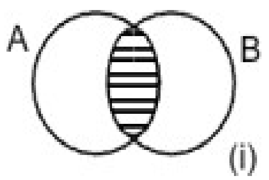
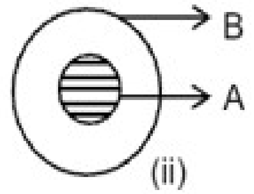
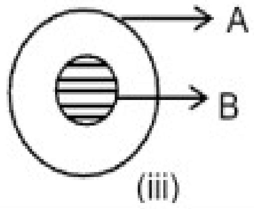
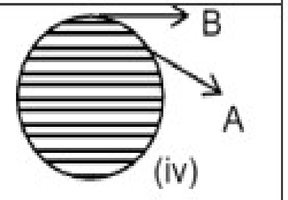
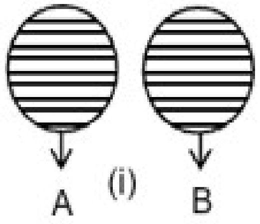
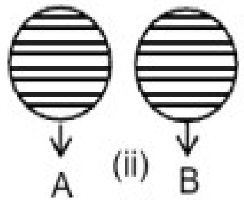
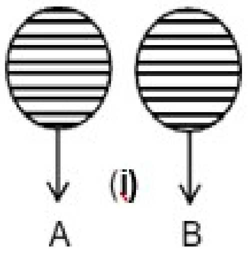
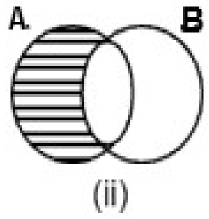
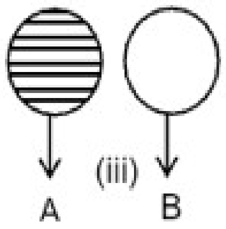
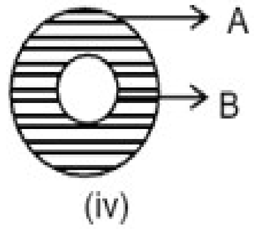
Qualifier	Alternate Forms
All A are B	Each A is B, Every A is B, Any A is B, A's are B
Some A are B	A few A are B, Many A are B, At least some A are B, Some A are definitely B
No A is B	None of the A are B, A can never be B, A is not B
Some A are not B	Some A can never be B, Some A are definitely not B, All A cannot be B

TYPE OF STATEMENTS

Type	Qualifiers Used
Affirmative	All, Some
Negative	No, Some-not



BASIC DIAGRAM & ALTERNATE DIAGRAM

	Basic Diagram (Minimum overlap)	Alternative Diagram		
(1) ALL: Eg: All A's are B's				
(2) SOME: Eg: Same A's are B's				
(3) NO: Eg: No A is B				
(4) SOME, NOT: Eg: Some A's are not B's				

1) **Statements:**

All hexagons are pentagons.
All pentagons are squares.

Conclusions:

I. All hexagons are squares.
II. All squares are hexagons.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II follow.

2) **Statements:**

All students are employees.
Some students are teachers.

Conclusions:

I. Some employees are teachers.
II. All teachers are employees.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II fallow.

3) **Statements:**

Some kids are males.
Some males are babies.

Conclusions:

I. All kids are babies.
II. No kid is a baby.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II follow.

4) **Statements:**

All ladders are mats.
All ladders are prisms.

Conclusions:

I. Some mats are prisms.
II. No mat is a prism.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II follow.

5) **Statements:**

All theorems are therapies.

Some therapies are sponges.

Conclusions:

I. No theorem is a sponge.

II. Some theorems are sponges.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II follow.

6) **Statements:**

All photos are poems.
No poem is a picture.

Conclusions:

I. Some photos are pictures.
II. No photo is a picture.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II follow.

7) **Statements:**

All kilobytes are megabytes.

Some kilobytes are not gigabytes.

Conclusions:

I. No megabyte is a gigabyte.

II. Some megabytes are not gigabytes.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II follow.

8) **Statements:**

All erasers are pencils.
All pens are pencils.
Some pens are sharpeners.

Conclusions:

I. Some erasers are pens.
II. Some pencils are sharpeners.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II follow.

9) Statements:

All employees are graduates.
Some graduates are students.
All students are guardians.

Conclusions:

I. At least some graduates are guardians.
II. Some employees are students.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II follow.

10) Statements:

All oceans are continents.
Some continents are countries.
No country is a lake.

Conclusions:

I. All lakes are continents.
II. Some lakes are not continents.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II follow.

11) Statements:

No king is a queen.
No queen is a prince.
No prince is brave.

Conclusions:

I. No king is prince.
II. Some queens are not brave.

A. if only I follows.
B. if only II follows.
C. if either I or II follows.
D. if neither I nor II follows.
E. if both I and II follow.

12) Statements:

Some cookers are machines.
Some machines are not ovens.
All ovens are televisions.

Conclusions:

I. Some televisions are definitely not machines.
II. All televisions are machines.
III. No cooker is oven.

A. Only III and either I or II follows.
B. Only I follows.
C. Both I and II follow.
D. Either I or II follows.
E. None follows.

13) Statements:

No holder is a hanging.
All berths are hangings.
Some hangings are tables.

Conclusions:

I. No holder is a berth.
II. At least some berths are tables.
III. Some tables can never be holders.

A. Both I and II follow.
B. Both I and III follow.
C. Both II and III follow.
D. All follow.
E. None follows.

14) Statements:

All dollars are rupees.
Only a few rupees are euro.
Some dollars are pound.

Conclusions:

I.Some rupees are pound
II. No euro is a dollars.
III. All pound are euro.

A. Only II follows.
B. Only I follows.
C. Only II and III follows.
D. All follows.
E. None follows.

15) Statements:

Some hot are cool.
All cool are warm.
Only a few hot are spice.

Conclusions:

I. Some cool are not spice.
II. Some hot are warm.
III. Some warm are cool.

A. Only I follow.
B. Only III follow.
C. Only I and II follow.
D. Only II and III follow.
E. Only I and III follow.

ANSWER KEY

1.A	6.B	11.D
2.A	7.B	12.B
3.D	8.B	13.B
4.A	9.A	14.B
5.C	10.C	15.D